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Received: 1 September 2025

Accepted: 2 May 2026

Published online: 20 May 2026

Cite this article as: Lou Z., Wang S., Yu X. *et al.* Predicting customer buying habits using convolutional neural network. *J Big Data* (2026). <https://doi.org/10.1186/s40537-026-01464-y>

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Predicting customer buying habits using convolutional neural network

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ABSTRACT

The key to personalized retail is to accurately predict consumer behavior, but traditional models can be problematic due to the large dimensionality of demographic data and non-linear relationships between demographics and behavior. In this paper, a new deep learning model is suggested, which applies a Convolutional Neural Network (CNN) to estimate the level of individual income and provide specific product suggestions. In contrast to the traditional tabular learners, our method converts normalized customer features to grayscale image matrices of size 20×10, allowing the CNN to learn the complex spatial features and latent behavioral patterns in the hybrid pooling layers. The algorithm is implemented in two combined steps: high-granularity income tier categorization and a recommendation engine that is powered by a purchase probability matrix. The experimental findings using a dataset of 980 people prove that the proposed model is much better than state-of-the-art benchmarks and has statistically significant accuracy of 93.06 in income prediction and 95 in recommendation success. These results highlight how the use of spatial feature extraction can be more effective in consumer analytics and offer a scalable pipeline to e-commerce real-time personalization.

Keywords: Convolutional neural network, Income prediction, Buying habits, Product recommendation.

1. INTRODUCTION

Customers are being prioritized in the business and have emerged as the dominant factor. With this in mind, firms must present customers with incentives in order to lessen the likelihood of them switching to competitors. A modest unpleasant encounter with a client may indicate that the customer may churn [1]. Many e-commerce use cases rely heavily on predicting future consumer behavior [2]. In other words, a company's customers are its lifeblood. Customers are at the center of marketing efforts, and organizations frequently make poor decisions when they ignore the behavior and motivations of their customers. Understanding the relationship between customer metrics and profitability and company value is crucial as marketing aims to become more accountable. There are many different types of customer metrics. It has been divided into two categories: perceptual and unobservable/behavioral metrics. Observable measurements comprise consumer behaviors that are usually associated with the acquisition or use of a good or service. Customer views (such as service quality), attitudes (such as customer satisfaction), and behavioral intents (such as purchase intention) are instances of unobservable constructs [3].

Market basket analysis, according to Russell and Petersen [4], focuses on the mechanism by which customers choose products from a certain set of groups within a single shopping trip. Its goal is to find relationships between the selections of various products made in a certain retail establishment, like a supermarket. According to Wang and Hong [5], shifts in consumer behavior lead to erratic customer profitability as well as wasteful and unproductive marketing strategy. In order to accomplish their marketing objectives, the authors present a customer profitability management system that makes use of data mining techniques. The development of the internet and e-commerce has completely changed how consumers make purchases as well as how companies or brands promote to draw in and keep customers by sending communications that are tailored to their individual needs [6].

The emergence of these e-commerce platforms has also made it more difficult for marketers and business owners to manage their operations effectively. In order to enhance purchase revenue, e-commerce will rely entirely on

technology and experts who can create customized shopping experiences for potential customers [7]. Due to this new trend's entire reliance on technology, there are numerous additional obstacles that must be overcome in order to realize the benefits of this new framework. Personalized options, the internet, appropriate product presentation, and many other aspects can all have an impact on the e-commerce business [8, 9].

It is crucial to predict income levels and comprehend the complex connection between income and purchasing patterns since it has impacts on both the commercial and the private spheres. From the perspective of business, such knowledge can be utilized for the improvement of marketing communication, for customization of products and increasing customer satisfaction. Thus, the segmentation of the target market according to the income level is very beneficial for the companies as it allows to invest more effectively and generate the highest possible revenues. On the same note, for individuals, a better understanding of spending behavior can help in making more informed decisions on their expenses and help in finding products and services that suit their needs and wants.

Moreover, the study of the correlation between purchasing behaviors and income also gives insights on consumers' behaviors. It enables the business to target the right audience and market its products and services to the right audience. For the researchers, the knowledge of this correlation can help to enhance the overall understanding of the economic drivers affecting consumers' decisions. It can also help to reveal information about tendencies of the society and inequality in consumers' behavior depending on income level.

Customer behavior analysis has been a process of identifying patterns in the data through techniques such as logistic regression. However, identifying relationships between data and analyzing patterns become complex issues especially when solving problems in large datasets. Convolutional Neural Networks (CNNs) are in fact a type of Deep Learning (DL) that can be applied to solve this problem effectively. CNNs are good at feature extraction which helps in their application for customer data analysis, and since customer data can be represented in the form of multi-dimensional arrays.

While numerous methods exist for predicting income and analyzing customer behavior, including traditional machine learning algorithms [10-19] and sales prediction/product recommendation approaches [20-26], many of these approaches suffer from limitations when dealing with the high dimensionality and complexity of modern datasets. Traditional machine learning methods together with basic neural networks face major challenges when attempting to identify the complex interactions between different customer characteristics, their earned income and purchase patterns. One of the greatest weaknesses of simple neural networks is that they cannot identify significant latent features of customer data in multi-dimension, which is a compounding of the problems that other conventional machine learning algorithms face with non-linear correlations between data. Basic neural networks demonstrate their ineffectiveness when processing complex data according to multiple research studies [27]. Machine learning Random forests show limitations when dealing with data sets that have very high dimensions [28]. The existing techniques need improvement because they fail to manage complex data requirements effectively.

Despite the fact that DL approaches are increasingly used for the analysis of customer behavior, there are no studies on the use of CNNs for the prediction of customers' attributes and purchasing behavior. Previous works could utilize a basic neural network architecture or other machine learning approaches besides deep learning. This study explicitly fills this gap by showing how CNNs are distinctly able to rise above these constraints. Compared to the traditional machine learning models and simple neural networks, which are not always able to capture complex feature interactions, CNNs have a greater ability to process high-dimensional structured data. In particular, using our proposed method, in which the standard one-dimensional demographic and behavioral data about customers are converted into two-dimensional 20×10 feature matrices, the CNN will be able to extract complex spatial correlations and latent patterns in a unique way [29]. This special spatial processing of tabular data enables the network to isolate very small correlations that earlier algorithms fail to capture, thereby greatly improving our accuracy in predicting income, and also making the results of our product recommendations very targeted.

Recent advances in deep learning [30-32], have shown promising results in various domains. The following research investigates the direct use of CNNs yet future work should consider integrating these approaches especially transfer learning methods to enhance model functionality. The mentioned strategies enhance model generalization while making it more resilient across different domains.

In our research, first, we endeavor to predict the income level using CNNs. Then, based on the projected income level, we aim to derive a more accurate model of individuals' purchasing habits. This, in turn, enables us to provide more precise recommendations. Our method not only predicts purchasing habits but also includes income level as a factor that has a significant impact on those habits, which gives a better view of the customers. The key contributions of the work are summed up as follows:

- Presenting a new architecture for CNN based on hybrid pooling layers for predicting people's income level.
- Presenting a probabilistic model for modeling the buying habits of customers and taking into account the income level.
- Using the combination of individual characteristics and customer behavioral records in buying products to form a recommender system.

The paper follows this progression: Similar works and backgrounds are examined in the second section. The introduced technique is explained in the third section, and the results obtained from its implementation are presented in the fourth section. Section 5 discusses the findings, implications, limitations and future works; and finally, the fifth section includes the conclusions.

2. Background and Related Works

The current research includes two areas, one is predicting people's income level and the other is recommending products. In this section, some recent researches have been studied in each of these categories.

2.1. Predicting People's Income Levels

Yamnampet [10] focused on income determination, using various classifiers to reduce cost and risk. It also demonstrates the performance of each algorithm in customer identification and analyzes scored probabilities, scored labels, false negatives, and true positives.

Thapa [11] evaluated the performance of five machine learning algorithms on an adult income dataset, revealing the Random Forest Classifier as the most effective with 86.3% training and 86% test accuracy.

Chen et al. [12] proposed random forest (RF) as a method for predicting salaries, demonstrating its superiority over traditional methods such as k-nearest neighbors, naive Bayes, logistic regression, and decision trees on the adult dataset. RF enhances accuracy through dataset preprocessing, variance reduction, and factor elimination.

Viroonluecha and Kaewkiriya [13] aimed to develop a system for predicting salary in Thailand based on Deep Learning, analyzing personal data from a job search website with over 1.7 million users. When compared to algorithms like Random Forest and Gradient Boost Trees, the model achieved an optimal R-squared result of 0.462.

Kablaoui and Salman [14] utilized a dataset of over 20,000 salaries in the USA to apply three supervised machine learning techniques: linear regression, random forest, and neural networks. It was found that the neural network outperformed the other models, achieving an accuracy of 83.2%.

Wang et al. [15] analyzed the factors influencing college graduates' starting salaries using machine learning methods at a 2020 financial university. The factors included academic qualifications, professional disciplines, employment regions, industries, gender, and student cadres, with the XGBoost model emerging as the best predictor.

Vemulapati et al. [16] discussed income prediction methodologies using Long Short-Term Memory (LSTM), ConvLSTM, Bi-Directional LSTM (BiLSTM), and Stacked LSTM networks. It highlighted pre-processing

approaches and training mechanisms, and demonstrated successful implementations. It was found that the BiLSTM was more accurate.

Chakrabarty and Biswas [17] presented an Income Prediction Model using Grid Search on Adult Census Data, Hyper-Parameter Tuning, Gradient Boosting, and Ensemble Learning, achieving a Validation Accuracy of 88.16%, the highest ever.

Rehman et al. [18] proposed to demonstrate the use of data mining and machine learning strategies to address income disparity issues. The study aimed to classify machine learning for predicting whether an individual has an annual income.

Wang [19] utilized 48,842 income census data from the Adult Data Set to predict annual income levels using machine learning approaches. Thirteen attributes, including native-country, occupation, marital status, education, workclass, and age, were considered. A randomly divided 32,561 individuals were trained using various algorithms.

2.2. Sales Prediction and Product Recommendation

Xian et al. [20] explored the use of historical sales and behavioral data analytics to create a recommendation model for online B2C businesses during the COVID-19 pandemic. They proposed a process model using RFM analysis, k-means clustering, and association rule theory for shopping basket analysis and product recommendations. This model improved sales, market responsiveness, and profitability for companies in similar situations.

Chaubey et al. [21] compared various machine learning techniques for predicting customer purchasing behavior. They utilized supervised classification methods such as dummy classifier, XgBoost, AdaBoost, ANN, SGD, random forest, SVM, Naïve Bayes, KNN, decision trees, and logistic regression. Additionally, they employed hybrid methods like SvmAda, RfAda, and KnnSgd. The best classification model was a hybrid method using an ensemble stacking technique (KnnSgd), achieving an accuracy of 92.42%.

Kumar et al. [22] introduced a new algorithm for predicting customer interest that used pattern mining techniques and Multi Variant K-means clustering. It identified user purchase histories, enquires, and purchase patterns, generating recommendations for advertisements and banner placement. This approach enhanced retail marketing strategies and customer retention.

Anitha and Patil [23] aimed to enhance business sales and profit by providing relevant and timely data on potential customers in the retail industry. The data was analyzed using a systematic approach employing the K-Means algorithm. The study used the Recency, Frequency and Monetary (RFM) model and dataset segmentation principles, validating various dataset clusters based on the Silhouette Coefficient.

Nguyen et al. [24] created a customized recommendation system based on a multi-stage retrieval approach. Their methodology is a combination of collaborative filtering, Bayesian Personalized Ranking (BPR) and popularity-based algorithms to produce candidate items, which are ranked with LightGBM and Deep Neural Networks (DNN).

Parihar and Yadav [25] investigated the use of machine learning to predict customer behavior in e-commerce by analyzing clickstream and customer data. Their goal was to use artificial intelligence to investigate output discrepancies in models, with a focus on sequential clickstreams and static consumer data, in order to increase customer loyalty and transactions.

Zhao and Keikhosrokiani [26] created a novel data science life-cycle and process model that uses RFM analysis and other analytics algorithms to anticipate sales and recommend products. They used customer segmentation and machine learning techniques to examine traditional shop business transformation. The prediction system, which used XGBoost and Random Forest, was able to forecast consumer orders with 77.82% accuracy. The recommendation system also used association rules to evaluate transaction statistics, revealing how online shopping platforms promote products to customers.

In order to give a clear picture of the existing research field and point out the gaps in the methodology that this study fills, Table 1 will present the summaries of the methodologies, goals, strengths, and limitations of the recent literature in the field.

Table 1. Summary of the related studies

Reference	Year	Methodology	Goal	Strengths	Limitations
Yamnampet [10]	2017	Various Machine Learning Classifiers	Income determination and customer identification.	Thorough evaluation of scored probabilities and error rates (false negatives/positives).	Relies on traditional classifiers; may struggle with highly complex, non-linear data interactions.
Thapa [11]	2023	ML Algorithms (Random Forest, etc.)	Adult income prediction.	Achieves strong baseline accuracy (86%) and provides a broad algorithm comparison.	Basic ML architectures lack the ability to extract deep spatial/latent features from datasets.
Chen et al. [12]	2022	Random Forest (RF)	Salary prediction on the adult dataset.	Enhances accuracy via rigorous preprocessing, variance reduction, and feature elimination.	Tree-based models can be limited when scaling to very high-dimensional behavioral data.
Viroonluecha & Kaewkiriya [13]	2018	Deep Learning vs. RF, Gradient Boosting	Salary prediction for Thailand labor workforce.	Utilizes a massive real-world dataset (CZAT users); DL outperformed traditional ML.	R-squared value of 0.462 indicates a significant portion of data variance remains unexplained.
Kablaoui & Salman [14]	2022	Neural Networks, Linear Regression, RF	Salary prediction on a USA dataset.	Demonstrates the superiority of basic neural networks (83.2% accuracy) over traditional ML.	Basic NNs are less effective than CNNs at capturing spatial relationships in structured data.
Wang et al. [15]	2022	XGBoost	Predict starting salary of college graduates.	High interpretability of demographic and academic influencing factors.	Highly domain-specific (financial university graduates); limited generalizability.
Vemulapati et al. [16]	2023	LSTM, ConvLSTM, BiLSTM, Stacked LSTM	Income prediction.	Effectively captures sequential data; BiLSTM showed high accuracy.	High computational complexity; RNN/LSTMs are less optimized for non-sequential spatial data than CNNs.
Chakrabarty & Biswas [17]	2018	Gradient Boosting, Ensemble Learning, Grid Search	Adult census income level prediction.	Achieves high validation accuracy (88.16%) through rigorous hyperparameter tuning.	Reliance on traditional tabular ML limits deeper feature extraction capabilities.
Rehman et al. [18]	2022	Data Mining, Machine Learning	Classify existence of annual income to address disparity.	Strong socio-economic application and focus on income inequality.	Focuses on binary/simple classification rather than granular income tier prediction.
Wang [19]	2022	Various Machine Learning Approaches	Predict annual income levels based on 13 attributes.	Comprehensive feature importance analysis on a standard dataset.	Constrained by the limitations of traditional ML in modeling complex behavioral matrices.

Reference	Year	Methodology	Goal	Strengths	Limitations
Xian et al. [20]	2022	RFM Analysis, K-means, Association Rules	B2C product recommendation during COVID-19.	Direct real-world business applicability; improves market responsiveness.	Static clustering approaches may fail to adapt to rapid, dynamic shifts in user behavior.
Chaubey et al. [21]	2023	ML & Hybrid Ensembles (e.g., KnnSgd)	Predict customer purchasing behavior.	High accuracy (92.42%) achieved through advanced ensemble stacking techniques.	Complex hybrid ensembles can be computationally expensive and lack model interpretability.
Kumar et al. [22]	2021	Pattern Mining, Multi Variant K-means	Customer interest prediction and ad recommendation.	Effectively uses purchase histories to optimize retail marketing strategies.	Clustering-based recommendations often suffer from the "cold-start" problem for new customers.
Anitha & Patil [23]	2022	K-Means, RFM Model	Enhance retail sales via customer segmentation.	Systematic validation of data clusters using the Silhouette Coefficient.	K-Means assumes spherical data clusters, which may not accurately represent complex human behavior.
Nguyen et al. [24]	2024	Retrieval Strategy (CF, BPR) + LightGBM/DNN	Improve user engagement via personalized product recommendations.	Successfully combines multiple algorithms to handle large-scale data and the cold-start problem.	Performance highly dependent on the retrieval of candidates; DNN components showed lower MAP@K compared to boosted trees.
Parihar & Yadav [25]	2021	Machine Learning	Predict consumer future preferences.	Successfully integrates both sequential clickstream data and static demographic data.	Output discrepancies require manual AI investigation, limiting end-to-end automation.
Zhao & Keikhosrokiani [26]	2022	XGBoost, RF, Association Rules, RFM	Sales prediction and product recommendation.	Provides a holistic life-cycle model combining both prediction and recommendation phases.	Accuracy of 77.82% suggests significant room for improvement via deep learning feature extraction.

2.3. CNNs with Hybrid Pooling

CNN models, despite their high efficiency in pattern learning, require a large training set. For small training sets, CNNs may face the problem of overfitting [33]. Overfitting reduces the generalizability of the CNN when applied to new instances. In many situations, the overfitting problem in CNNs can be attributed to the function used in the pooling layers, based on which one can obtain the feature map extracted from the data through the activations output of the convolutional layers [34]. In CNN designs today, two pooling algorithms are frequently utilized. The first function is called max pooling, and it takes each feature map region's maximum activation value. This allows for the extraction of the most prominent features and the elimination of less significant ones. In real-world applications, this function causes overfitting [35].

Average pooling, the second operator, takes into account the results for a region in equal measure. The ReLU activation and average pooling operator work together to lessen the impact of strong activations while simultaneously producing a significant amount of zeros in the feature map. However, when average pooling is combined with other activation operators—like the hyperbolic tangent function—it may cause loss of data by ignoring positive as well as negative activations. In light of these drawbacks, numerous research have introduced novel pooling techniques in an effort to reduce the overfitting issue in CNNs. Hybrid pooling [36] can be an

efficient solution. It combines the benefits of both max pooling and average pooling, thereby improving the generalizability of CNNs.

This strategy seeks to boost the adaptability of the CNN model by leveraging diverse pooling strategies during training and averaging their predictions at test time. During training, each feature map in the convolution layer undergoes both average pooling and max pooling. The choice between them is randomized, governed by a probability (p) for average pooling and $(1-p)$ for max pooling. This approach, as described in [36], essentially blends the benefits of both pooling methods, aiming to achieve superior generalizability compared to relying on a single strategy. This mechanism can be formulated as follows [36]:

$$S = \begin{cases} S_{avg} & \text{with probability } p \\ S_{max} & \text{with probability } 1 - p \end{cases} \quad (1)$$

Where S_{avg} is the output of average pooling operator for various regions and is defined by the set $S_{avg} = \{s_{avg}^1, \dots, s_{avg}^J\}$. The following relationship holds for each member of this set:

$$s_{avg}^j = \frac{1}{|R_j|} \sum_{i \in R_j} a_i \quad (2)$$

where, R_j shows the j th pooling segment, which includes a group of activations such as $\{a_1, \dots, a_{|R_j|}\}$. On the other hand, in eq. (1), S_{max} refers to the output of max pooling operator for various segments and is defined by the set $S_{max} = \{s_{max}^1, \dots, s_{max}^J\}$. For each member of this set, the following relationship holds:

$$s_{max}^j = \max_{i \in R_j} a_i \quad (3)$$

Then, in the test phase, the result of any pooling area is measured using eq. (4) [36]:

$$S = S_{hybrid} = p \times S_{avg} + (1 - p) \times S_{max} \quad (4)$$

This approach seeks to enhance CNN model diversity by strategically combining two distinct pooling operators for various feature maps.

3. Research Methodology

This section provides a detailed description of the data collection process. It also outlines the steps of the proposed method for predicting customer purchasing habits. This method leverages deep learning techniques.

3.1. Data

The data used in this research was collected through a comprehensive questionnaire, which was meticulously designed to capture a wide range of variables. The gathered information through each questionnaire from participants include demographics, shopping habits, income level, and product preferences of the participants. The used questionnaire and response options have been presented in Appendix A. This questionnaire was distributed among 980 individuals, ensuring a diverse sample in terms of age, gender, and income levels. The 980 participants are well balanced in terms of demography: 48 of them were males and 52 of them were females. Age categories were divided into five with the highest proportion of 25-34 (31.10 %) and 35-44 (27.56%) representing the largest proportion of the workforce consumers. Geographically, respondents were mostly located in a wide range of urban and suburban areas to capture a wide range of retail accessibility and economic conditions. The responses were then organized into a structured dataset for further analysis. This dataset includes detailed information about the purchasing habits of individuals across six different categories of goods.

The categorization of goods in this study includes mobile phones, tablets, laptops, clothing, entertainment items, and food. For each category, a set of brands available at the data collection site was identified, and respondents

were asked to indicate their preferences. In addition to the information on people's purchasing habits, a table was created to describe the characteristics and behavior of each individual. Table 2 provides a comprehensive overview of the collected information for each individual in this dataset.

Table 2. List of descriptive information for each individual.

Row	Feature	Type
1	Gender	Nominal
2	Marital Status	Nominal
3	Age	Numeric
4	Education Level	Ordinal
5	Job Category	Nominal
6	Job History	Numeric
7	Place of Residence	Nominal
8	Residential Status	Ordinal
9	Specific Disease	Nominal
10	Alcohol Consumption	Nominal
11	Consumption of Tobacco or Other Addictive Substances	Nominal
12	Number of In-person Shopping Instances per Week	Numeric
13	Number of Online Shopping Instances per Week	Numeric
14	Quantity of cart during In-person Shopping	Numeric
15	Quantity of cart during Online Shopping	Numeric
16	Number of Working Hours per Week	Numeric
-	Income Level	Numeric

Based on the input information for the income level feature, respondents are divided into five categories of income levels: very low (164 samples), low (178 samples), average (208 samples), high (193 samples), and very high (237 samples). The goal of the proposed method is to predict individuals' income levels based on the independent features listed in Table 2. Then, modeling the purchasing habits based on the determined income levels.

Although the data used in this study has a solid basis, some weaknesses in the data collection process can be identified. The use of a questionnaire opens the possibility of self-reporting bias especially on sensitive variables like precise income or shopping frequency. Moreover, the sampling was also geographically limited to certain areas, which can affect the extrapolation of the buying patterns to other cultural or rural settings. These were alleviated by careful design of the questionnaire and the weighted cross-entropy loss function to make the model resilient to different samples.

3.2. Proposed Method

It seems that a person's habits and preferences during shopping are related to their behavioral characteristics and income level. Therefore, this research attempts to model people's shopping habits based on the information obtained from their income classification. The proposed method models customer behavior and recommends goods in three steps:

1. Preprocessing,
2. Classification of individuals based on income
3. Product recommendation based on income.

The structure of the presented approach is illustrated in Figure 1. First, the collected information from individuals (Table 2) is normalized and transformed so that it can be processed by learning models. Then, in the second step, the individual's features are processed by a CNN, and their income level is predicted to determine the target category for the individual based on the predicted value.

In the third step of the presented approach, a suitable product is recommended based on the predicted income level. At this stage, a purchase probability matrix is utilized to model people's shopping habits based on their income level. This matrix, which has a number of rows equal to the number of recommended categories for goods and a number of columns equal to the number of income groups, models the probability of purchasing a product in a specific category by individuals in different income groups.

As illustrated in Figure 1, the data collected via the questionnaire is divided into two categories: training data and test data. The dataset instances are permuted randomly and then partitioned into subsets using the cross-validation approach (as explained in section IV) to determine the training and test sets. Initially, in the training phase, the purchasing habits model, denoted as P , is formed based on the purchase histories and preferences of training instances. Then, for each sample belonging to the test set, they are first classified based on their income level using the CNN model. Based on the predicted income and the probability matrix P , a product is recommended to the individual.

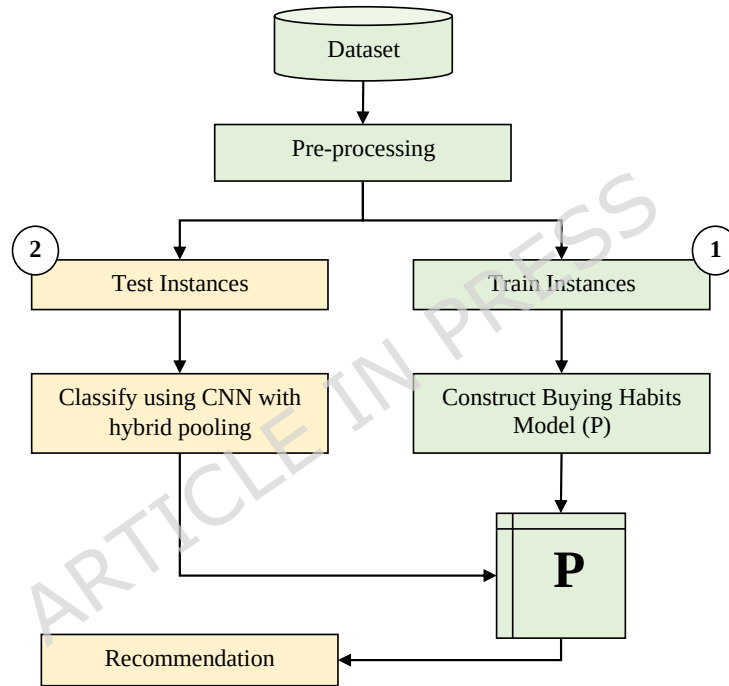


Figure 1. Structure of the presented approach.

3.2.1 Preprocessing

The proposed method utilizes a straightforward mechanism for preprocessing the data, which encompasses three sub-steps: 'value conversion', 'missing value management', and 'feature normalization'. The preprocessing begins with the conversion of nominal values into numerical ones. In this process, each value in the nominal features is converted into a natural number. For ordinal nominal features, a unique list is compiled based on the rank of the values present in that feature. Subsequently, a value like IX is assigned to each value in the sorted list, proportional to its position. This approach ensures a streamlined and effective preprocessing of the data. For non-ordinal nominal features, a list of unique values is created based on the frequency of each nominal value. Ultimately, each value in the mentioned feature is replaced with a natural number corresponding to it in the IX set. By executing the above process, all features of the dataset are converted into a numerical format.

The choice of this particular numerical mapping, as opposed to more traditional numerical encoding methods, including direct one-hot encoding or target encoding, is specifically meant to be compatible with the high-dimensional feature-to-image mapping in Section 3.2.2. Although standard encoding works well with more traditional tabular learners, it usually creates sparse vectors which do not provide the structure density needed by convolutional filters. In our design, this preliminary mapping will give a standardized number base that will enable all the features to be divided into 100 fine-grained intervals. This generates the 1600-bit binary representation that is required to make a complete 20×10 grayscale image. Through the use of this approach instead of traditional approaches, we are able to have the CNN perceive demographic and behavioral characteristics as geometrical patterns, and the model is able to identify deep, non-linear relationships between disparate characteristics that would have been missed by standard encoding schemes.

Subsequently, records with missing values are corrected. For this purpose, if a feature with a missing value in a record is numerical, it is replaced with the mean of the existing values for that feature. Conversely, for nominal features with missing values, the missing value is replaced with the mode or the value with the highest frequency for that feature.

The preprocessing step concludes with feature normalization. During this process, the value vector of each feature, denoted as x , is mapped to the range $[0,1]$ using eq. (5).

$$N_x = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (5)$$

Where x_{min} and x_{max} respectively describe the smallest and largest values for feature (x).

3.2.2 Classification of Individuals Based on Income Using CNN

In this step, a CNN with hybrid pooling layers is employed to classify individuals based on their income level. For this purpose, a matrix representation of the set of normalized features obtained from the previous step is used. The process of converting tabular customer data to a format that can be processed by convolutional processing consists of four steps of discretization and mapping:

1. Interval Mapping: Each normalized feature $x \in [0, 1]$ is mapped to one of 100 equal intervals of length 0.01.
2. High-Dimensional Encoding: Using one-hot encoding, each interval is converted into a binary string of length 100. As an example, a value between $[0.01, 0.02)$ will give a 1 at the second position and 0 at other positions.
3. Bit-Stream Aggregation: With each record being represented by a high-dimensional bit-stream of 1600 bits (16×100), all 16 features are concatenated to generate a bit-stream.
4. Spatial Grayscale Mapping: This bit-stream is divided into blocks of 8 bits with each block being mapped to a single pixel intensity (0-255). The pixels are then reformed into a 20×10 grayscale image matrix.

This organized representation enables the CNN to process customer profiles as spatial patterns to enable the latent feature correlations that are usually misplaced in traditional vector-based models to be identified. The obtained matrix is fed to a CNN classifier. The CNN model uses this input to predict the individual's income level.

The reason for selecting CNNs for this research is their ability to efficiently handle large, structured data, such as the matrix representation of individual characteristics employed in our study. Their hierarchical organization, incapacity for translation, and abilities in parallel processing qualify them for activities that involve pattern recognition and classification in such data. When compared with additional deep learning methods like Recurrent Neural Networks (RNNs) or LSTMs, CNNs can deliver higher performance in tasks dedicated to spatial relations and high-dimensional data. In contrast, RNNs and LSTMs are mainly created for sequential data, in which the

arrangement of elements is important. Though they can comply with the data organization detailed in this study, they may not be as efficient as CNNs in capturing spatial features.

In order to increase the generalization capabilities of the model on small and imbalanced data, we apply the Hybrid Pooling strategy described in Section 2.3. This mechanism contrasts with max and average pooling by switching between max and average pooling stochastically during training unlike in the case of static pooling. This helps the model to avoid over-fitting to the most salient activations (which is one of the main causes of overfitting in customer behavior data) and to make sure that the feature maps do not lose the subtle behavioral indicators. Figure 2 shows the particular arrangement of these layers into our architecture.

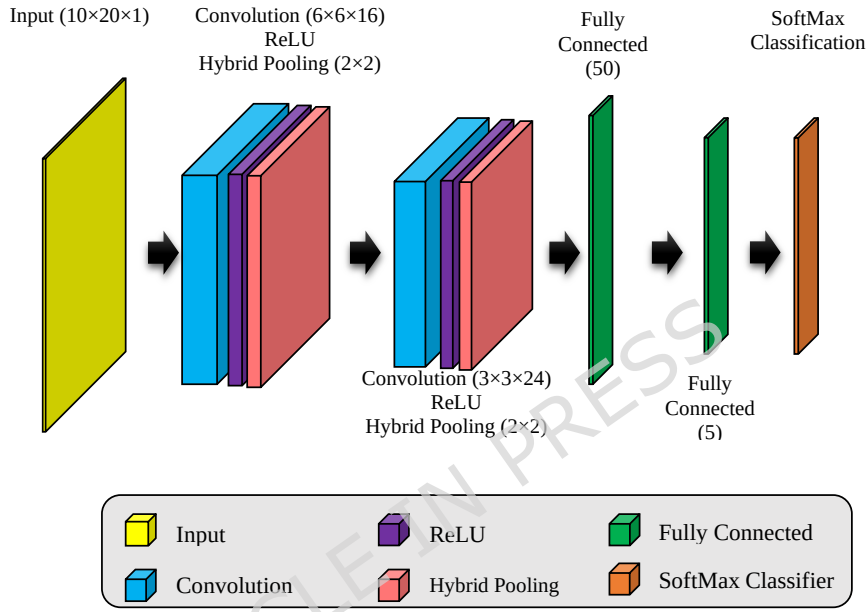


Figure 2. The proposed CNN model for predicting individuals' income levels.

Another major issue with income prediction is the imbalance in classes (164 very low vs. 237 very high samples). In order to solve this without losing useful data, we use a Weighted Cross-Entropy (WCE) loss function. The technical explanation behind WCE is to impose a penalty weight w_i on each of the classes inversely proportional to its frequency in the training set. This compels the gradient descent algorithm to make decisions that are more focused on the proper classification of minority groups, which practically eliminates the majority bias that is usually rife with conventional CNNs. The WCE loss is mathematically formulated as:

$$Loss = -\sum (w_i * y_i * \log(p_i) + (1 - y_i) * \log(1 - p_i)) \quad (6)$$

In Eq. (6), w_i is the weight of the class i and for this class its true label is y_i . Furthermore, p_i is the probability that is being predicted for the class i .

The careful tuning of hyperparameters is essential to optimize the performance of CNN models. The optimal hyperparameter setting for the employed CNN structure was obtained using the grid search strategy. In this regard, various hyperparameter settings of the CNN were examined using the training loss metric. The examined hyperparameters for tuning the model include the dimensions and number of convolutional filters, dimensions of the pooling layers, and also type of activation layers. Additionally, various settings for training-related parameters of mini batch size and optimizer were considered in the tuning step. Table 3 shows the search space for each configurable parameter of the CNN in this research.

The best obtained configuration of the CNN has been presented in Fig. 2. The inputs of the introduced CNN are defined as the matrix that described at the beginning of this section. This CNN comprises two layers of convolution with 16 and 24 filters, respectively. The width and length of these layers are 6×6 and 3×3 , respectively. The output of each of these layers are processed through the ReLU activation function, and feature map extraction at each stage is performed by hybrid pooling with dimensions of 2×2 . For both convolution and pooling layers, the stride was considered as 1. Finally, two consecutive fully connected layers are utilized to extract features. The first one transforms the extracted features into a vector form. The second fully connected layer calculates the probability of the sample belonging to each of the target categories. This is done by outputting a posterior probability vector. Ultimately, these features are classified by a SoftMax layer to predict the income level for the sample based on it.

Table 3. the search space for each configurable CNN parameter

CNN parameter	Search space
Dimension of convolution layers	{2, 3, 5, 7, 9}
Number of convolution filters	{4, 8, 16, 24, 32, 48}
Dimension of pooling layers	{2, 3, 4, 5}
Activation function	ReLU, PReLU, Leaky ReLU
Optimizer	SGDM, Adam
Mini batch size	{16, 32, 64}

3.2.3 Modeling Shopping Habits and Recommending Goods Based on Income

After employing the proposed CNN model for classifying individuals based on their income level, a matrix is used to model people's shopping habits and offer suggestions to them based on the created model. This matrix shows which goods people with different incomes are more likely to buy and which category of goods they are less likely to purchase. The structure of the probability matrix used in the proposed method is shown in Figure 3.

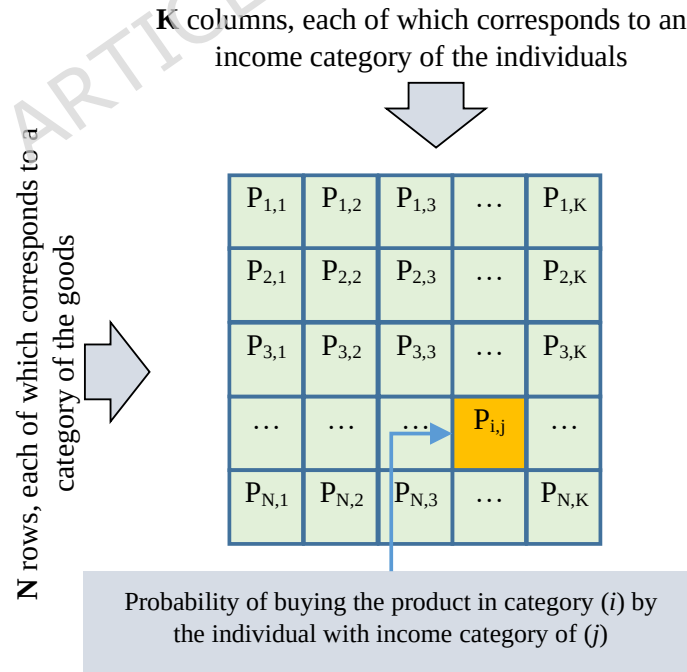


Figure 3. The structure of the probability matrix for modeling shopping habits and recommending goods to individuals in the proposed method.

Referring to Figure 3, the depicted matrix is structured with its rows equivalent to the recommendable goods and its columns corresponding to the distinct income groups. Consequently, this probability matrix is characterized by N rows and K columns. Each matrix element specifies the probability that an individual, belonging to the income group represented by the current column, will purchase the goods associated with the current row category. In essence, $P_{i,j}$ in Figure 3 represents the probability of an individual from income category j purchasing a good from category i . During the data collection phase, each participant provided personal information and responded to queries about their purchasing habits for various goods. Consequently, each individual's purchase record can be denoted as $\langle N_i, K_i \rangle$, where N_i signifies the category of goods that the individual purchased, and K_i designates the income group to which the individual belongs. Therefore, the database can be structured as depicted in Table 4, for each category of recommended goods.

Table 4. Data structure for storing individuals' shopping habits in the proposed method.

Row	Income Group	Purchased Goods Category
1	K1	N1
2	K2	N2
...	...	...
X	KX	NX

In Table 4, X denotes the number of records in the database. To calculate the probability matrix presented in the proposed method, first, the total selection of each category of goods by each income group of individuals is calculated as follows:

$$M_{i,j} = |\{ \langle N_c, K_c \rangle | N_c = i \text{ and } K_c = j \}| \quad (7)$$

Where $1 \leq c \leq X$ is the counter for the database records. In the above relation, the number of records where an individual in category j has chosen a good in category i is counted. After calculating the matrix M , each element located in the matrix M is divided by the sum of the elements of the column related to that element.

$$P_{i,j} = \frac{M_{i,j}}{\sum_{k=1}^N M_{k,j}} \quad (8)$$

By applying the above relation to each element of matrix M , the probability matrix P is obtained, which indicates the probability of different groups of individuals choosing different categories of goods. After forming the matrix P , the act of suggesting goods to individuals can be performed. For this purpose, after receiving the personal information of each person, the classification operation is first performed using the CNN model. In the next step, if the individual is classified into class (j) , the good with the highest probability in the (j) -th column of the matrix (P) is suggested to the individual.

4. Research Finding

The proposed approach was implemented and evaluated using MATLAB 2020. We reviewed the presented approach in two scenarios: in the first scenario, we predict people's income level, and in the second phase, based on the model that was considered to predict people's buying habits, we make recommendations.

In this study, stratified 5-fold cross-validation (CV) was utilized for the proposed method. First, we permuted the dataset instances randomly and then, divided the data into 5 parts and each part made up 20% of the data set. Then we repeated the operation 5 times. During each repetition, 80% of the data was used for training (based on 70% of instances) and validating (based on 10% of the instances) the model. After training the model, the remaining 20% unseen data is used as test data, and the classification result for each test sample is compared with its ground-truth label, leading to the one of the following conditions:

- True Positive (TP): The quantity of positive cases accurately detected by the model.

- False Negative (FN): The quantity of cases that the model incorrectly identified as negatives.
- False Positive (FP): The quantity of samples in which the model incorrectly classified as positive.
- True Negative (TN): The quantity of samples in which the model properly detected a negative.

Accuracy represents the overall proportion of correct predictions made by a model. Mathematically, this can be expressed as:

$$Accuracy = 100 \times \frac{TP+TN}{TP+FP+TN+FN} \quad (9)$$

Prioritizing true positives, precision measures the percentage of genuine positives among all model-predicted positives. Crucial for minimizing false positives. Mathematically:

$$Precision = \frac{TP}{TP+FP} \quad (10)$$

Focusing on completeness, recall measures the percentage of true positives accurately identified out of all actual positive instances. Crucial for avoiding missed positives. Mathematically:

$$Recall = \frac{TP}{FN+TP} \quad (11)$$

F-Measure provides a balanced assessment of both precision and recall. It is often employed when both metrics hold equal importance. Mathematically:

$$F - Measure = \frac{2*Precision*Recall}{Precision+Recall} \quad (12)$$

4.1. Evaluation in terms of Predicting People's Income Level

We evaluated the proposed method in this phase based on individuals' income. This method was assessed using the criteria of f-measure, recall, precision, and accuracy, and we compared it to other methods.

We evaluated the recommended method in three different modes: the proposed method, CNN (Max.pool), and CNN (Avg.pool). Due to the use of hybrid pooling layers that combines average and maximum pooling layers, we compared our proposed CNN model with two other CNN models that only use max pooling and average pooling layers. Also, we compared our proposed method with references [17-19], which we named Chakrabarty & Biswas, Rehman et al and Wang respectively.

Figure 4 shows the average accuracy, precision, recall, and f-measure graphically. In Figure 4a, our method demonstrates superior performance in predicting individual income levels, exceeding both the CNN (Max.pool) and Wang methods in average accuracy by margins of 1.5% and 2.7% respectively. This clearly indicates the improved accuracy of our method. In Figure 4b, our method outperforms the CNN (Max.pool) and Wang methods in precision by 1.5% and 2.6% respectively, indicating fewer false positive predictions and higher precision in identifying specific income levels. Additionally, our method surpasses Wang's method by a significant 2.8% in recall, demonstrating its superior ability to identify relevant instances within a dataset, including specific income levels. Furthermore, the 1.6% increase in f-measure compared to the CNN (Max.pool) method highlights the superior balance between precision and recall offered by our proposed method. In conclusion, our proposed method excels in accurately predicting individual income levels, as evidenced by its superior f-measure, recall, precision, and accuracy metrics.

The proposed CNN architecture takes the advantage of both average and max pooling, while attempting to overcome their limitations using the hybrid pooling layers. Average pooling aggregates all the feature activations in a given region and max pooling identifies the most active region. Using hybrid pooling layers, makes our model to more efficiently extract features from the customer data than models that employ one type of pooling only. This is likely to have been the reason for the enhanced performance as depicted by Figure 4.

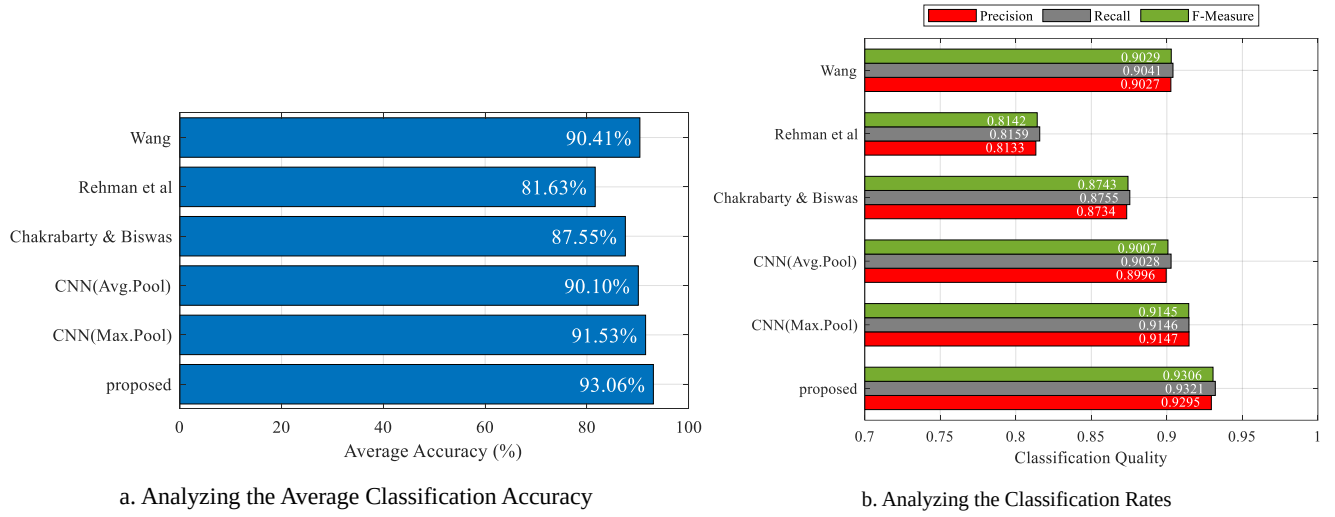


Figure 4. Evaluating the quality of the classification.

The proposed method achieves the highest values across all metrics: precision 0.9295, recall 0.9321, F-measure 0.9306, and accuracy 93.06%. Following closely are CNN (Max.Pool) with precision 0.9147, recall 0.9146, F-measure 0.9145, and accuracy 91.53%, and CNN (Avg.Pool) with precision 0.8996, recall 0.9028, F-measure 0.9007, and accuracy 90.10%. The methods Chakrabarty & Biswas, Rehman et al., and Wang exhibit lower performance, with Rehman et al. demonstrating the lowest values across all metrics.

Figure 5 presents an illustration of the confusion matrices pertaining to the introduced methodology and alternative comparative methodologies. Within this matrix, the rows depict the ground-truth classification of the instances, while the columns refer to the output of each model, arranged in alphabetical order. The elements situated along the main diagonal signify the accurate classification of the samples, whereas the non-main elements signify the quantity of classification errors found in the test samples.

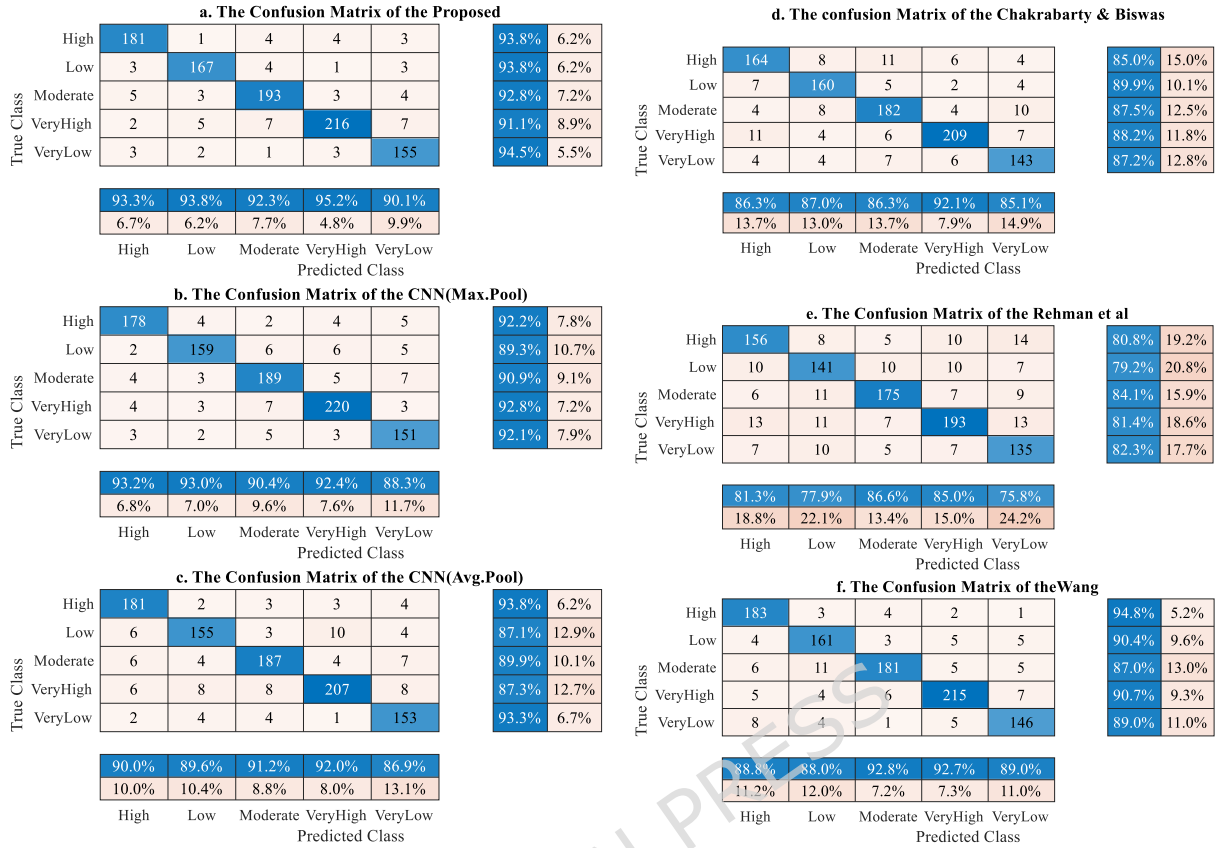


Figure 5. Evaluating the confusion matrix.

By segregating our confusion matrix from other confusion matrices, such as image 5b and image 5c, it becomes evident that our method, utilizing hybrid pooling layers, exhibits superior performance in segregating each group of data compared to the CNN (Max.pool) and CNN (Avg.pool) layers. Additionally, our matrix reveals that our proposed methodology has demonstrated better performance in segregating categories compared to the confusion matrices (5d, 5e, and 5f). The confusion matrix for our proposed method in figure 5a showed that the elements in the diagonal are higher than the other methods, meaning more number of elements correctly classified income levels. Apart from this, the off-diagonal elements (errors) are also less in our matrix as compared to the other methods which clearly explains that the proposed model is efficient enough to classify the individuals into their respective income category.

The proposed method yields the highest accuracy of 93.06% among all the methods under comparison. Our method yields the lowest value of false positive rate, but at the same time, it yields a higher true positive rate for each of the income groups, which is evidence of the fact that the improvement in accuracy is not only due to the reduction of false positives, but due to the increase of true positives as well.

Figure 6 depicts the ROC curve, presenting the FPR (false positive rate) and TPR (true positive rate) values for evaluating our multi-class classification. We computed the TPR and FPR for each class separately. After obtaining these TPR and FPR values, we aggregated and compared the ROC curve results for each method across all classes. The ROC curve for our proposed method exhibits a larger area under the curve compared to other approaches. This signifies a superior balance between the TPR and FPR achieved by our model. In simpler terms, our method effectively identifies a higher proportion of true income levels (high TPR) while minimizing the number of false positive classifications.

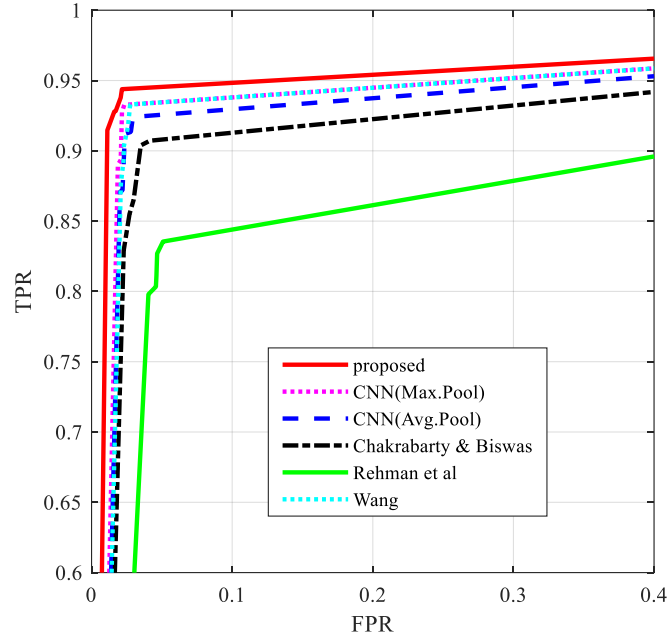


Figure 6. Evaluating the ROC curve.

The methods presented by Chakrabarty & Biswas [17], Rehman et al. [18], and Wang [19] might face issues in terms of modeling the interactions of the features within the customer data due to the model complexity or inability to handle imbalanced datasets in which certain income brackets could be under-represented in the dataset. On the other hand, the proposed CNN with hybrid pooling has a more flexible architecture that can potentially capture a wider range of interactions between the features.

Table 5 compares the performance of various methods using precision, recall, F-measure, and accuracy metrics. In this table, in addition to previously compared approaches, the RF [12] and BiLSTM [16] models have been considered for comparisons. For a fair comparative analysis, all of the models were trained and tested using the same instances. The proposed method achieves the highest values across all metrics. Following closely are CNN (Max.Pool) and CNN (Avg.Pool) confirming the effectiveness of CNNs in solving this problem, even while using conventional pooling layers. The BiLSTM [16] and RF [12] models show performance close to the proposed method. This indicated the desirable performance of deep learning and ensemble learning approaches in predicting income level. The methods Chakrabarty & Biswas [17], Rehman et al. [18], and Wang [19] exhibit lower performance, with Rehman et al. demonstrating the lowest values across all metrics.

Table 5. Performance comparison between the proposed method and previous approaches in predicting income level.

Methods	Precision	Recall	F-Measure	Accuracy
Proposed	0.9295	0.9321	0.9306	93.0612
CNN (Max.Pool)	0.9147	0.9146	0.9145	91.5306
CNN (Avg.Pool)	0.8996	0.9028	0.9007	90.1020
Chen et al. [12]	0.9077	0.9084	0.9079	90.9184
Vemulapati et al. [16]	0.9125	0.9108	0.9111	91.0204
Chakrabarty & Biswas [17]	0.8734	0.8755	0.8743	87.5510
Rehman et al. [18]	0.8133	0.8159	0.8141	81.6327
Wang [19]	0.9027	0.9041	0.9029	90.4082

The methods [17-19] might face issues in terms of modeling the interactions of the features within the customer data due to the model complexity or inability to handle imbalanced datasets in which certain income brackets could

be under-represented in the dataset. On the other hand, the proposed CNN with hybrid pooling has a more flexible architecture that can potentially capture a wider range of interactions between the features.

4.2. Statistical Significance Analysis

As explained in Section 4.1, stratified 5-fold cross-validation was used to strictly assess the performance of the proposed framework. The findings of the suggested CNN model were contrasted with two architectural variants (Standard Max Pooling and Average Pooling) and five models of the recent literature [12, 16, 17, 18, 19] and the results were reported in Table 5. To show that the performance gains obtained by the proposed approach are not the result of chance, a statistical significance analysis was performed.

Table 6 provides the summary of the average accuracy, standard deviation, and the paired t-test results. The highest mean accuracy of 93.02% and the lowest standard deviation (0.53%), which was obtained with the proposed model, not only proves high performance but also demonstrates high stability of the model on various data subsets.

Table 6. Comparison of Statistical Analysis of Model Accuracy (5-Fold CV)

Model	Average Accuracy	Std. Dev. Acc.	p-value
proposed	93.0612	0.5332	0.00000
CNN(Max.Pool)	91.5306	0.8396	0.00506
CNN(Avg.Pool)	90.1020	1.0290	0.00076
Chen et al. [12]	90.9184	0.5773	0.00372
Vemulapati et al. [16]	91.0204	0.7287	0.01563
Chakrabarty & Biswas [17]	87.5510	1.9962	0.00576
Rehman et al [18]	81.6327	1.5962	0.00003
Wang [19]	90.4082	0.6981	0.00049

As shown in Table 6, all calculated p-values are significantly lower than the standard alpha level ($\alpha = 0.05$). The fact that our hybrid pooling method is significantly better than standard CNN pooling mechanisms ($p < 0.01$) confirms that our hybrid approach to pooling does not cause the loss of spatial information in feature extraction to the same extent as standard CNN pooling mechanisms. Moreover, the fact that our 20×10 spatial mapping strategy outperforms the established benchmarks like Vemulapati et al. [16] ($p = 0.015$) and Wang [19] ($p < 0.001$) shows that our strategy is more effective at capturing latent socio-economic correlations compared to the traditional deep learning or machine learning architecture. These results are strong statistical indicators that the suggested changes to the CNN pipeline bring a substantial improvement in customer behavior analytics.

4.3. Evaluations in terms of product recommendation

In this phase, we have investigated the efficiency of the presented model from the aspect of product recommendation. In other words, when we provide a recommendation for product types, we compared that product with customers' buying habits and based on this comparison, we estimated the accuracy using eq. (8). The results of the proposed approach have been compared with several models including CNN with Max Pooling layers for (instead of hybrid pooling), and two baseline approaches: LightGBM [24], and RFM [26].

In Figure 7, the mean accuracy of product classification is illustrated. Our method has achieved a mean accuracy of 95% in category 1, and also reached average accuracies of 92.89% and 94.81% in categories 4 and 6, respectively. These results demonstrate that our approach has performed very well compared to the comparative method RFM and the CNN (Max. Pool) method.

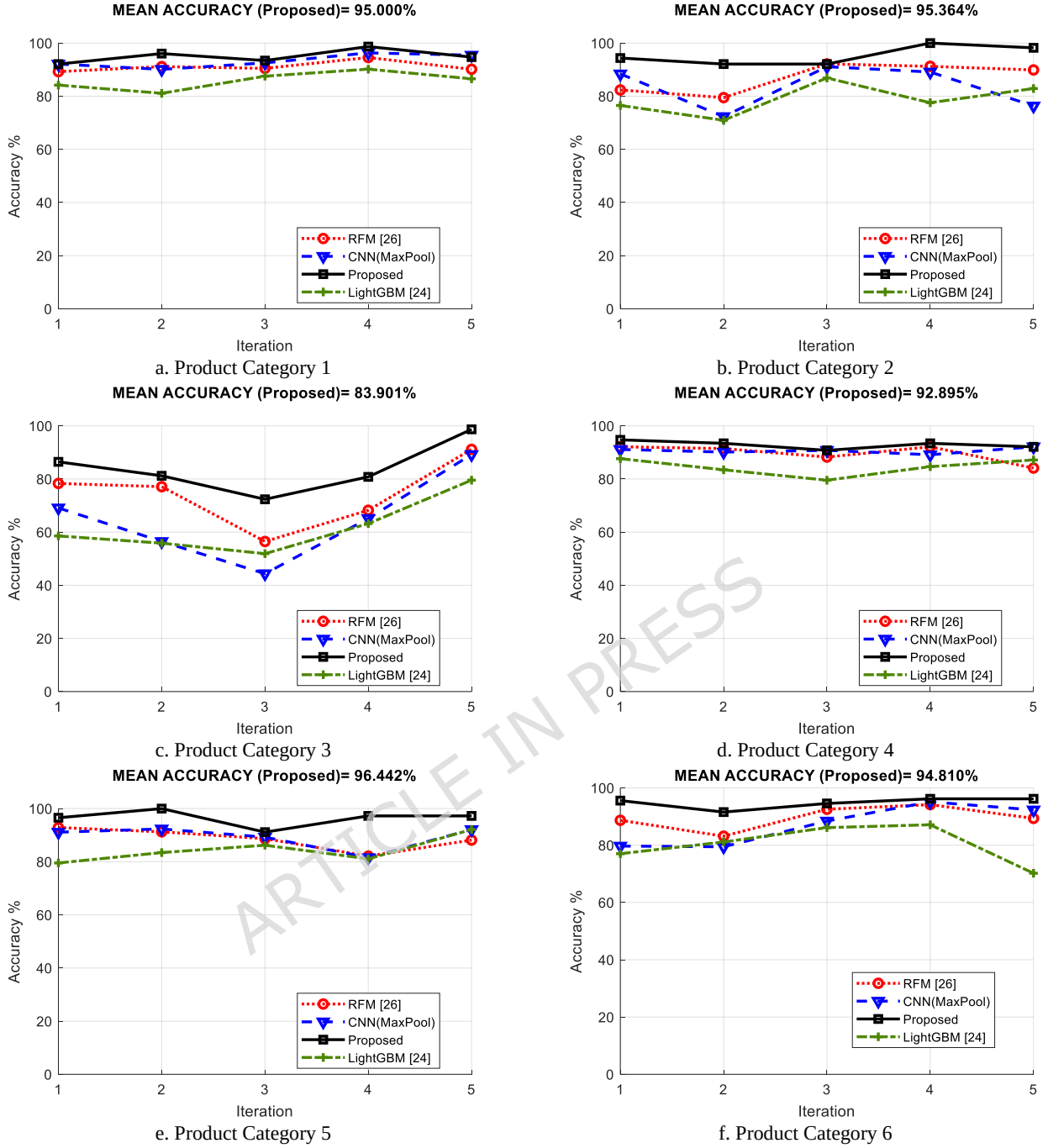


Figure 7. Evaluating the product categories accuracy.

4.3. Feature Importance Analysis

This experiment analyzes the importance of various features and their contribution in prediction of income level by the employed CNN structure. To do this, an attention layer was added between the input layer and the first convolutional layer of the network. This layer, assigns weight to each input feature based on their importance and transfers the weights to the subsequent layers. Since in input feature was translated to a binary vector of length 100, the obtained weights for all 100 bits of each feature were summed to obtain the overall weight of the attribute. After that, the obtained sum of weights for all features were normalized using the max-min approach (Equation 1). Figure 8 demonstrates the obtained normalized weight for each input feature which reflects its importance in predicting income level of the individuals.

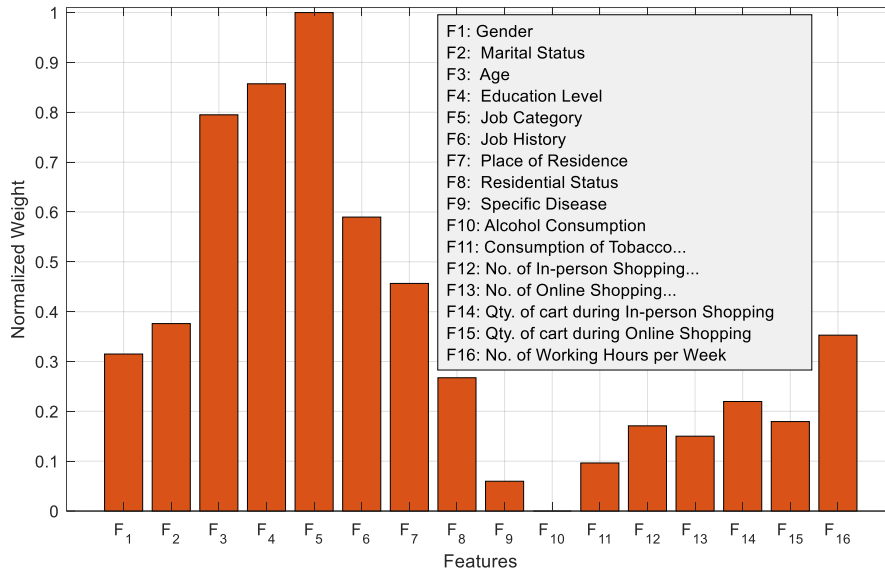


Figure 8. The normalized importance of features obtained through employing an attention layer in the structure of the CNN

As it is depicted in Figure 8, the six most important variables that can be useful in determining the income level include “Job Category”, “Education Level”, “Age”, “Job History”, “Place of Residence”, and “Marital Status”.

It is common knowledge that job category is usually a pointer to income levels, particularly in the emerging markets. There is normally a positive relationship between the level of education and the income earner’s wages or salaries. The third dependent variable age is also taken into consideration as it defines the career level and further more earning capability in hierarchical corporate environment. Whereas, stable & progressive job history has possibility that only career advancement and higher earnings might be expected. Additionally, there is a higher probability of higher average income in the urban areas especially the tier 1 and tier 2 cities. Finally, some cultures may determine economical and financial aspects of a couple’s financial life based on marital status and therefore this factor, based on income level, is the sixth in the list. However, it should be pointed out that the significance of these factors may differ based on the cultural, economic and social environment of the place from which the data was collected.

4.4. Practical Implications and Real-World Applications

The suggested CNN-based pipeline and probability matrix has a high potential of practical implementation, especially in e-commerce and retail ecosystems. When used to classify users in real-time as an income level by deploying the CNN model, platforms can automatically classify users into income levels based on initial demographic or session data. After classifying one into an income group, the system uses the purchase probability matrix (P) to customize user interfaces (UI) by ranking high-probability product categories in search results and recommendation banners. This architecture can support dynamically tuned marketing policies; an example would be to offer a 'High' income user premium electronics or luxury goods and offer a 'Low' income user value-oriented alternatives. This kind of focused strategy does not only improve the customer experience, but also helps retailers to manage their inventory effectively. By forecasting the demand of the income-specific products, businesses can optimize the stock levels and minimize the overhead expenses and the chances of stockouts.

In the financial services industry, the model can be applied to evaluate the credit status of individuals thus minimizing on defaults. Besides, it can also assist financial institutions in providing relevant financial services including loans and investment products according to the customer’s financial strength.

When applied to the public policy area, the model can be used to determine who among the population requires social help and therefore those who should be given the resources. Furthermore, the model can be used by policymakers to understand the distributional effect of economic policies in order to make informed decisions.

Finally, in the area of consumer rights advocacy, the model can be applied to search for cases of unjustified or misleading pricing or promotion of products for low-income consumers. Also, through the knowledge of the consumer behavior and their choice, the advocacy groups can then call for the production of safer and cheaper products.

With the help of using the AI and machine learning possibilities in processes the suggested model can foster fair and progressive society.

5. Discussion, limitation, and Future Works

The current study aimed at assessing the outcomes of a new CNN model with the incorporation of hybrid pooling layers for estimating customer income levels and then using it for the product recommendation tasks. The results show that the proposed method outperforms the previous methods in the accuracy, precision, recall and F1-measure. This section provides a further discussion on the significance of these findings, how these findings expand the knowledge of consumers' behavior, and possible applications. We also present the limitations that were experienced during the research and recommend future research directions.

The success of our CNN architecture in predicting income levels, achieving an overall accuracy of 93.06%, suggests its ability to capture complex relationships within customer data. By incorporating income level as a key factor, our approach offers a more nuanced understanding of how income demographics influence buying habits. This advancement goes beyond traditional methods that might struggle to capture these intricate interactions. The proposed method gives insight into the possibility of using CNNs with hybrid pooling in the analysis of customer behavior. To validate the generalization ability of the proposed model we used 5-fold cross validation. This evaluation methodology ensured that there was no overfitting of the models since it was quite a rigorous process. Comparison between hybrid pooling and the other forms of pooling such as max pooling and average pooling showed that the former outperformed the latter in terms of accuracy, precision, recall and F1-score. These results decisively indicate that the employment of the hybrid pooling strategy positively contributes to the improvement of the model's capability to generalize from unseen data. The use of average and max pooling through hybrid pooling layers in our model allows for the extraction of more complete features from customer data as compared to the use of standard pooling alone.

This finding contributes to the growing body of research exploring the effectiveness of deep learning techniques in understanding consumer behavior, especially considering the significant improvement of 2.7% in accuracy compared to the Wang method (90.4%) which utilizes conventional pooling layers. The fact that one can forecast the income levels and make recommendations based on this data (mean accuracy of 93.1% for product recommendations) is quite practical in reality.

The possibility of precise income level forecasting and the factors influencing consumers' behavior are significant for different industries. For instance, the information in this context can be used by retailers to manage their stocks, design and implement marketing strategies, and design new products that suit the needs of their target customers. This can have a positive impact on customer satisfaction, the conversion rate of sales, and better resource utilization. Such information may help to evaluate the credit standing of clients and adjust the offered financial services. In addition, such models can be useful for the policymakers in order to improve the economic policies and fight against the income disparity. Aside from the practical implications, our research contributes to the literature on consumer behavior and the part that income plays in consumers' decisions. It can be useful for the public policy debate on issues such as income disparity and consumer rights.

5.1. Limitations and Future Works

Even though the proposed model is highly accurate and practically useful, certain limitations should be considered. These are divided into limitations of the present and future research directions.

Research Limitations:

- **Population Representativeness:** Although the 980-sample sample is all-inclusive in this study, it might not be an accurate reflection of the entire world population, which can create predictive biases in highly different socio-economic settings.
- **Geographic and Cultural Specificity:** The present results are based on the regional data; hence, the performance of the model can differ in the case of other cultural backgrounds with their own shopping patterns.
- **Feature Engineering Scope:** The existing feature set, although strong, might not have necessary granular cultural or micro-economic indicators that can be used to determine individual buying choices.
- **Internal vs. External Validation:** Cross-validation has been done to ensure that the model is highly generalized in the given dataset, but the stress-testing of the model against completely external datasets that are independent of the given dataset has not been conducted yet.

Future Research Directions:

- **Increasing Data Sources:** Future research must use a variety of data sources across different countries to enhance the external validity and globalizability of the results.
- **Adding Longitudinal Data:** By adding variables that track time-series, the model would be able to adjust to changing consumption trends as time progresses.
- **Advanced Architectures:** We plan to explore Transformer-based models and transfer learning techniques to further enhance sequence modeling and reduce the data training requirements for unfamiliar domains.
- **Interdisciplinary Integration:** Cooperation with behavioral scientists to incorporate psychological profiling may offer a better insight into the reason behind the expected buying behavior.
- **Hardware and Efficiency Optimization:** Investigating memristive CNN architectures could significantly improve computational and energy efficiency, enabling real-time processing for large-scale industrial applications.

6. Conclusion

The paper focused on the utilization of a convolutional neural network to analyze and model customer purchasing habits in relation to their income levels. The study consisted of three primary stages: data preprocessing, categorizing individuals based on their income levels, and providing product recommendations tailored to their income brackets. The primary aim of the paper was to enhance understanding of individuals' purchasing behaviors and to improve the accuracy and relevance of product recommendations personalized to their specific income levels. The results indicated that the presented approach outperformed alternative techniques by increasing average accuracy to 93.06% and precision to 92.95%, thus demonstrating superior performance in predicting individuals' income levels. Additionally, the proposed method achieved at least 2.77% improvement in terms of f-measure compared to the benchmark methods. This significant improvement shows the high quality of the classification results produced by the proposed method. Furthermore, in the phase 2, our method showed that it has a mean accuracy of 95% which is at least 4.05% higher compared to other comparative methods in product recommendation.

APPENDIX A

This appendix provides a comprehensive overview of the questionnaire used for data collection and includes the exact wording of questions and response options.

Demographics

1. Gender:
 - Male
 - Female
 - Prefer not to say
2. Marital Status:
 - Single
 - Married

- ☐ Divorced
- ☐ Widowed
- ☐ Living with Partner
- ☐ Prefer not to say
- 3. Age: (Please enter your age)
- 4. Education Level:
 - ☐ High School Diploma or Equivalent
 - ☐ Associate's Degree
 - ☐ Bachelor's Degree
 - ☐ Master's Degree
 - ☐ Doctorate or Professional Degree
 - ☐ Prefer not to say
- 5. Job Category: (Please select the category that best describes your current job)
 - ☐ Management/Professional
 - ☐ Sales/Service
 - ☐ Skilled Trades/Labor
 - ☐ Administrative/Clerical
 - ☐ Student/Unemployed
 - ☐ Retired
 - ☐ Other (Please specify): _____
- 6. Job History (Years): (How many years have you been working in your current field?)
- 7. Place of Residence:
 - ☐ Urban Area (City)
 - ☐ Suburban Area
 - ☐ Rural Area
- 8. Residential Status:
 - ☐ Own Home
 - ☐ Rent Apartment/House
 - ☐ Live with Family/Friends
 - ☐ Other (Please specify): _____
- 9. Do you have any specific chronic diseases? (Yes/No)
 - ☐ If yes, please specify: _____

Shopping Habits

- 10. Alcohol Consumption:
 - ☐ Never drink alcohol
 - ☐ Drink occasionally
 - ☐ Drink regularly
- 11. Consumption of Tobacco or Other Addictive Substances: (Yes/No)
 - ☐ If yes, please specify: _____
- 12. On average, how many times do you shop in person per week?
- 13. On average, how many times do you shop online per week?
- 14. On average, how many items do you typically add to your cart during in-person shopping trips?
- 15. On average, how many items do you typically add to your cart during online shopping trips?
- 16. On average, how many hours per week do you typically work?

Income Level

- 17. We understand that income level can be a personal question. If you are comfortable sharing this information, please select your income range:
 - ☐ Below \$25,000
 - ☐ \$25,000 - \$49,999
 - ☐ \$50,000 - \$74,999
 - ☐ \$75,000 - \$99,999
 - ☐ \$100,000 and above
 - ☐ Prefer not to say

Product Preferences

Thank you for participating in this survey! This section focuses on your preferences for various product categories. Please note: There is no right or wrong answer. We are simply interested in understanding your brand choices.

If you don't have a preference for a particular category, you can skip that section.

Mobile Phones:

- Brand 1: _____
- Brand 2: _____
- Brand 3: _____ (or "No Preference")

Tablets:

- Brand 1: _____
- Brand 2: _____
- Brand 3: _____ (or "No Preference")

Laptops:

- Brand 1: _____
- Brand 2: _____
- Brand 3: _____ (or "No Preference")

Clothing:

- Brand 1: _____
- Brand 2: _____
- Brand 3: _____ (or "No Preference")

Entertainment Items:

- Brand 1: _____ (e.g., TVs, Video Game Consoles)
- Brand 2: _____
- Brand 3: _____ (or "No Preference")

Food:

- Brand 1: _____ (e.g., Grocery Stores, Restaurants)
- Brand 2: _____
- Brand 3: _____ (or "No Preference")

Funding

The work described in this paper was supported by a grant from Social Science Foundation of Shandong Province, China (Grant No. 17CGLJ15). This article is the research outcome of a university-level project at Shandong Women's University (Project Approval Number: 2021RCYJ02/57).

Data availability

All data generated or analysed during this study are included in this published article.

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